Milestone 2: Scientific Contextualization

Priority Inheritance Protocol (Resource Access Protocols)

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I. Protocol Landscape and Chronology

Chronological development (Buttazzo §7.1 overview):

1) **NPP** (Non-Preemptive Protocol): classical; no cited inventor. Task runs non-preemptively inside CS. Simple, over-blocks.
2) **HLP/IPC** (Highest Locker Priority): classical. Raises priority at entry time to ceiling of entered semaphore. Requires ceiling declarations in code.
3) **PIP + PCP**: Sha, Rajkumar, Lehoczky [SRL90], IEEE Trans. Computers 39(9):1175–1185, 1990. *Same paper, two protocols*: PIP (§7.6) and PCP (§7.7) as a design progression.
4) **SRP**: Baker [Bak91], Real-Time Systems 3(1):67–99, 1991. Extends PCP: multi-unit resources, EDF, stack sharing.

Conceptual positions:

- NPP/HLP block at *activation time* (before CS entry). Pessimistic.
- PIP defers blocking to *access time* (at `pi_wait`). No blocking unless actual contention. Less pessimistic.
- PCP adds lock-grant rule on top of PIP: prevents deadlock and chained blocking. Non-transparent.
- SRP blocks at *preemption time* (like NPP/HLP) but with tighter dynamic ceilings; allows EDF and stack sharing.

II. Where PIP Sits Relative to Alternatives

PIP vs. NPP: NPP blocks all higher-priority tasks whenever *any* CS is entered, even when those tasks share no resource with the CS task. PIP blocks only tasks that actually contend for the same semaphore. More selective, less pessimistic.

PIP vs. HLP: HLP blocks a task at *activation time* based on ceiling of semaphore being entered — even if the branch accessing the resource is never taken at runtime (conditional-CS pessimism, Buttazzo p. 214). PIP defers to actual `wait` call. Removes conditional-branch pessimism. However, HLP gives single blocking (Theorem 7.1) vs. PIP's α_i blockings (Theorem 7.2) — HLP can be *less* blocking in worst case.

PIP vs. PCP:

Property	PIP	PCP
Deadlock-free	**No**	Yes (Thm 7.3)
Chained blocking	**Possible**	No (Thm 7.4)
Transparent	Yes	**No**
Max blockings	$\alpha_i = \min(l_i, s_i)$	1
Blocking time	Tighter (actual contention)	Pessimistic (ceiling rule)
Impl. effort	Hard	Medium

PIP vs. SRP (Baker [Bak91]): SRP extends PCP with: (1) multi-unit resources, (2) EDF support via preemption levels, (3) runtime stack sharing (up to 90% savings, Buttazzo §7.8.5). PIP is fixed-priority, single-unit, no stack sharing.

III. Kopetz: Priority Inversion as Correctness Failure

Kopetz definition (RTS slide 7): "A real-time computer system is one in which correctness depends not only on logical results but also on the *physical instant* at which they are produced."

Implication: In the Fig. 7.4 scenario, τ_1 has highest priority because its deadline is tightest. If τ_1 misses its deadline due to unbounded inversion, the physical instant of its output is wrong — not just late. This is a system failure by definition, not a QoS degradation.

Slide 13 (Prof. Henkler): "Real-time means predictable; result after deadline is wrong, not just late."

Slide 48: EDF optimal *only without resource competition.* PIP (or equivalent) is necessary to restore theoretical scheduling guarantees in the presence of shared resources.

IV. Application Domains

POSIX (`PTHREAD_PRIO_INHERIT`):

- POSIX.1-2008 standardises `pthread_mutexattr_setprotocol` with `PTHREAD_PRIO_INHERIT`.
- One attribute change at mutex creation; zero changes to lock/unlock sites.
- Supported: Linux (with RT patch / kernel $\geq$2.6.25), VxWorks, QNX.
- **Mars Pathfinder (1997):** VxWorks mutex without priority inheritance caused IPC bus watchdog resets. Fix: enable `PTHREAD_PRIO_INHERIT` on the meteorological bus mutex. One config change, no code rewrite.

Automotive OSEK/AUTOSAR:

- OSEK OS default: ceiling protocol (HLP/IPC variant), not PIP.
- Some AUTOSAR vendor extensions and POSIX-compliant RTOS layers provide PIP-like inheritance (**qualified claim**: not a universal AUTOSAR requirement).
- ISO 26262 (ASIL): bounded blocking time required for safety qualification. PIP's B_i bound + RTA provides this.

V. Where PIP Is Insufficient

Multiprocessor systems:

- Ch. 7 explicitly uniprocessor (§7.3, p. 209).
- Boosting a holder's priority on a different processor requires inter-processor interrupt (IPI) with non-deterministic latency.
- Gracioli et al. [GP] measure IPI overhead (Δ_{ipi}), cache affinity loss, scheduling overhead on actual hardware — none appear in Buttazzo's B_i formula.
- **Consequence:** PIP semantics cannot be directly applied on multiprocessors; MPCP / FMLP / MSRP needed.

Distributed systems: Resource access becomes message-passing; priority inheritance requires network communication; timing is non-deterministic. Kopetz [Ko] discusses distributed timing constraints; PIP provides no help here.

EDF scheduling: PIP defined for fixed priorities. For EDF systems, SRP [Bak91] is the standard protocol (handles dynamic priorities via preemption levels).

VI. Scientific and Engineering Tradeoffs

Tradeoff	PIP advantage	PIP disadvantage
Pessimism vs. correctness	Lower pessimism: blocks only on genuine contention	Multiple blockings (α_i) vs. single blocking in HLP/PCP
Transparency vs. safety	No code change needed; legacy-friendly	Dynamic inheritance hard to audit statically; limits formal certification
Simplicity vs. completeness	No deadlock risk from protocol itself	Programmer must enforce total semaphore ordering
Generality vs. scope	Works for any fixed-priority RTOS	Uniprocessor only; no EDF support

VII. Reference Justifications

Buttazzo 2011: Primary source. Provides all formal definitions, lemmas, theorems, algorithms, and the five-protocol comparison.

Sha, Rajkumar, Lehoczky 1990: Original PIP/PCP paper [SRL90]. Required to cite original authors of Theorem 7.2 and PCP results.

Kopetz 2011: Grounding definition for real-time correctness (physical instant, slide 7). Establishes why inversion is a correctness failure, not a performance issue.

Abella et al. 2015: WCET estimation limits. Contextualises the assumption that $\delta_{i,k}$ values are exactly known. None of the surveyed WCET methods is fully trustworthy.

Gracioli et al. 2013: Multiprocessor OS overhead. Grounds PIP's scope limitation with measured data from real hardware.

Baker 1991: SRP origin [Bak91]. Required for protocol landscape (Table 7.5 SRP row) and distinguishing EDF-capable protocols.

Behrmann et al. 2004: UPPAAL tutorial. Needed for M4 UPPAAL implementation and verification methodology.

Excluded references (with justification):

- General scheduling textbooks (not PIP-specific; Buttazzo covers scope).
- AUTOSAR specifications (not publicly available; claim qualified instead).
- POSIX.1-2008 standard (too broad; PTHREAD_PRIO_INHERIT is a well-known fact, not requiring a primary citation in a 5-page paper).